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Q1) A What is phishing? Explain the step-by-step algorithm to identify phishing emails.

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1. Definition of Phishing

Phishing is a type of cyberattack in which an attacker sends fake emails or messages pretending to be a trusted organization, such as a bank, to steal sensitive information like passwords, bank details, OTPs, and credit card numbers.

Example: An attacker sends an email saying, “Your bank account will be blocked. Click here to update your account details.” The link opens a fake banking website that steals the user's information.

2. Step-by-step algorithm to identify phishing emails

Algorithm: Identify_Phishing_Email

Step 1: Start.

Step 2: Receive the email from the sender.

Step 3: Check the sender's email address. If the address is suspicious or different from the official address, mark it as suspicious.

Step 4: Check the subject line. If it contains urgent or threatening words such as “Account Blocked,” “Urgent Action,” or “Verify Now,” increase suspicion.

Step 5: Check the email content for spelling mistakes, grammatical errors, and requests for passwords, OTPs, or banking details.

Step 6: Check the links in the email. If a link leads to an unknown or suspicious website, mark the email as suspicious. Do not click the link.

Step 7: Check attachments. If an unexpected attachment is received, especially an executable file, do not open it.

Step 8: If multiple suspicious signs are found, classify the email as a phishing email. Otherwise, classify it as not identified as phishing and continue to exercise caution.

Step 9: Report the suspicious email to the email service provider or IT administrator.

Step 10: Stop.

4. Common signs of phishing emails

1. Unknown or fake sender address.
2. Urgent or threatening language.
3. Suspicious links or fake websites.
4. Spelling and grammatical mistakes.
5. Requests for passwords, OTPs, or bank details.
6. Unexpected or suspicious attachments.

Conclusion: Phishing emails can be identified by carefully checking the sender, subject, message content, links, and attachments. Users should never share confidential information through suspicious emails.

Q2) B Write a simple Python program to hide a text message inside an image and display the hidden message again.

```
from PIL import Image
import stepic

# Function to hide message
def hide_data():

    # Ask user for original image
    image_name = input("Enter the original image name: ")

    # Open image
    img = Image.open(image_name)
```

```
    # Ask user for secret message
    secret_msg = input("Enter the secret message: ")

    # Convert message into bytes
    data = secret_msg.encode("utf-8")

    # Hide message inside image
    stego_img = stepic.encode(img, data)

    # Ask for output image name
    output_name = input("Enter name for stego image (use .png):
")

    # Save as PNG
    stego_img.save(output_name, format="PNG")

    print("Message successfully hidden!")
    print("Stego image:", output_name)


# Function to retrieve message
def retrieve_data():

    # Ask user for stego image
    image_name = input("Enter the stego image name: ")

    # Open stego image
    stego_img = Image.open(image_name)

    # Decode hidden data
    decoded_data = stepic.decode(stego_img)

    # Convert bytes to normal text
    print("Retrieved Message:", decoded_data.decode("utf-8"))

# Main program
hide_data()
retrieve_data()
```

Or

Q2) B Write a Python program to simulate a phishing login page for educational purposes..

```
print("================================")
print(" LOGIN PAGE")
print(" EDUCATIONAL SIMULATION")
print("================================")
# Dummy login details
correct_username = "student"
correct_password = "12345"
# Take input from user
username = input("Enter Username: ")
password = input("Enter Password: ")
# Check login details
if username == correct_username and password ==
correct_password:
print("\nLogin Successful!")
print("This is only a simulation.")
else:
print("\nLogin Failed!")
print("Invalid Username or Password.")
# Security message
print("\n--- Security Awareness ---")
print("Do not enter your password on unknown websites.")
print("Always check the
```